

Breast cancer

Every decision runs off **subtype** (HR+/HER2−, HER2+, TNBC) crossed with **stage** · AJCC 8th adds grade + ER/PR/HER2 as **prognostic staging** (does not apply pathologically after neoadjuvant therapy)

Stanford Oncology Handbook 2025, Breast Cancer (Schultz C, Khan S, Sidhu K, Yang M, Dalal N, Tsai J) · NCCN 5.2025 · items marked † are outside the handbook chapter

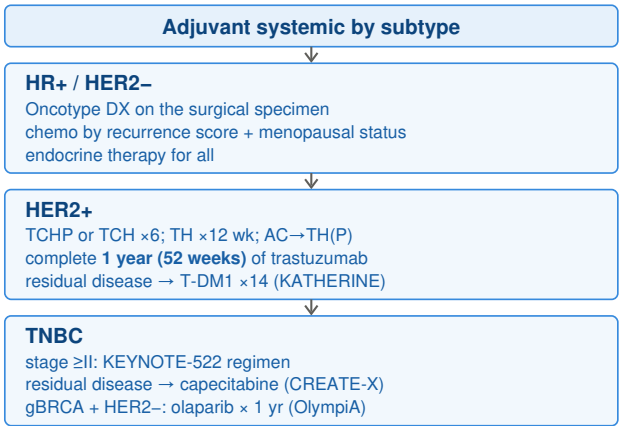
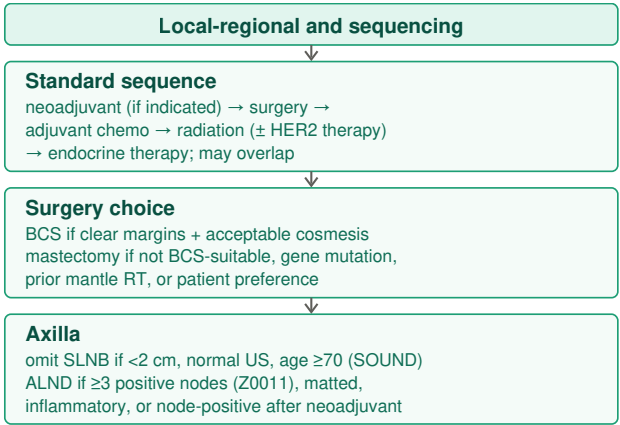
WORKUP

- Bilateral mammography + ultrasound; MRI or contrast-enhanced mammogram if indicated
- Core biopsy with **clip placement**
- Pathology: ER/PR, **HER2 (IHC ± DISH — must do DISH if IHC 2+)**, grade, Ki-67
- “High risk”: HR weak/negative, HER2+, TNBC, high grade, **Ki-67 >20%**, inflammatory, angiosarcoma, malignant phyllodes
- **Genetic testing**: dx age **≤50 (NCCN)** or **≤65 (ASCO)**, bilateral at any age, or TNBC
- **Stage I–IIB: no routine staging imaging** — only if directed by signs, symptoms or labs
- Stage III: CT chest/abdomen ± pelvis + bone scan, or PET/CT
- Stage IV or recurrent: brain MRI if CNS symptoms, spine MRI if cord-compression symptoms, and **biopsy the first recurrence** — ideally a metastatic site

STAGING, MARGINS & IN-SITU DISEASE

T1 / T2 / T3	≤20 mm / >20–50 mm / >50 mm
T4	chest wall (4a) · skin edema, ulceration, satellite nodules (4b) · both (4c) · 4d = inflammatory
pN1 / pN2 / pN3	1–3 / 4–9 / ≥10 axillary nodes ; pN1mi = >0.2–2 mm; pN0(i+) = isolated tumor cells
Negative margin	≥2 mm for DCIS and invasive; positive or close → discuss re-excision
Post-neoadjuvant	prefix yc / yp ; pathologic complete response = ypT0 ypN0 cM0
DCIS	lumpectomy + whole-breast RT · or mastectomy ± SLN · or lumpectomy alone if small and age >70 . If ER+, tamoxifen × 5 yr (NSABP B-24)
LCIS	florid / pleomorphic → excise to negative margins; classical does not need negative margins

CURATIVE-INTENT PATHWAY



NEOADJUVANT THERAPY — LOCAL-REGIONAL DISEASE

Subtype	Neoadjuvant regimen	If residual disease
HR+ / HER2−	ddAC → T, or TC ×4 — the same regimens used adjuvantly	continue endocrine therapy ± CDK4/6 escalation
HER2+	TCHP ×6 (preferred), or AC → THP	T-DM1 3.6 mg/kg q21d ×14 (KATHERINE)
TNBC, stage ≥II	KEYNOTE-522 : carboplatin + paclitaxel + pembrolizumab → AC + pembrolizumab	capecitabine ×6–8 (CREATE-X); if gBRCA, olaparib × 1 yr

Indications: inflammatory or inoperable disease · T4 or N2–3 · TNBC or HER2+ with T2 or N1 · to downsize for breast conservation or downstage the axilla. **T2/T3 meeting BCS criteria except for size** → consider neoadjuvant. Response guides post-operative escalation — the main argument for treating first.

† ONCOTYPE DX — WHEN TO ADD CHEMOTHERAPY

Setting	Recurrence score → recommendation
Node-negative, any age	RS 0–10 → endocrine therapy alone
Node-negative, age >50	RS 11–25 → endocrine therapy alone; chemotherapy adds nothing
Node-negative, age ≤50	RS 16–25 → consider adding chemotherapy — a real benefit in this group
Node-negative, any age	RS ≥26 → chemotherapy + endocrine therapy
1–3 nodes, postmenopausal	RS ≤25 → endocrine therapy alone — no chemotherapy benefit
1–3 nodes, premenopausal	RS ≤25 → chemotherapy still benefits
1–3 nodes, any	RS >25 → chemotherapy + endocrine therapy

Node-negative cutoffs from **TAILORx** (Sparano, NEJM 2018); node-positive from **RxPONDER**. Applies to **ER/PR-positive, HER2-negative** disease with up to 3 positive nodes; sent on the surgical specimen. † **The handbook does not state these cutoffs** — it defers to NCCN.

ADJUVANT ESCALATION — HR+/HER2−

Agent	Regimen and eligibility
Abemaciclib (monarchE)	150 mg BID × 2 years with endocrine therapy — ≥4 positive nodes , OR 1–3 nodes plus grade 3, tumor ≥5 cm, or Ki-67 ≥20%
Ribociclib (NATALEE)	400 mg daily, 21 days on / 7 off × 3 years with endocrine therapy — any stage IIB or III, or stage IIA with grade 3 (or grade 2 with high genomic risk or Ki-67 ≥20%)
Olaparib (OlympiA)	300 mg BID × 1 year — high-risk HER2-negative disease with a germline BRCA1/2 pathogenic variant

Breast cancer

REGIMENS & DOSING

Regimen	Dosing	Cycle
ddAC → T	doxorubicin 60 + cyclophosphamide 600 d1 → then paclitaxel 175 (or 80). Filgrastim d3–10 or pegfilgrastim d2	AC q14d ×4; then T q14d ×4 or q7d ×12
TC	docetaxel 75 + cyclophosphamide 600 d1	q21d ×4
KEYNOTE-522	carboplatin AUC 5 d1 + paclitaxel 80 d1, 8, 15 + pembrolizumab 200 d1 → then doxorubicin 60 + cyclophosphamide 600 d1	q21d ×4 each; pembrolizumab × 1 year
Capecitabine	1000–1250 mg/m ² PO BID d1–14	q21d ×6–8
Trastuzumab	4 → 2 mg/kg weekly, or 8 → 6 mg/kg q3wk	complete 52 weeks
TCHP / TCH	docetaxel 75 + carboplatin AUC 6 + trastuzumab 8→6 mg/kg (± pertuzumab 840→420) d1	q21d ×6
TH	paclitaxel 80 weekly ×12 + trastuzumab 4→2 mg/kg weekly, then 6 mg/kg q3wk to 1 year	q7d ×12
AC → TH(P)	AC q14d ×4 → paclitaxel 80 weekly ×12 + trastuzumab ± pertuzumab to 1 year	AC q14d ×4
T-DM1	3.6 mg/kg IV d1 — residual disease after preoperative therapy	q21d ×14

All doses mg/m² unless stated. **Avoid concurrent trastuzumab — especially trastuzumab + pertuzumab — with an anthracycline:** significant cardiotoxicity. Cardiac monitoring every 3 months on trastuzumab. If T-DM1 is stopped for toxicity, give trastuzumab ± pertuzumab to complete 1 year.

BONE-DIRECTED THERAPY

Setting	Agent and dose
Bone metastases	zoledronic acid 4 mg IV q4wk (may extend to q12wk after 12 months) · pamidronate 90 mg IV q3–4wk · denosumab 120 mg SQ q4wk
Osteoporosis prevention	denosumab 60 mg SQ q6 months · zoledronic acid 5 mg IV yearly · alendronate 70 mg PO weekly · calcium 1000 mg + vitamin D 800 IU daily

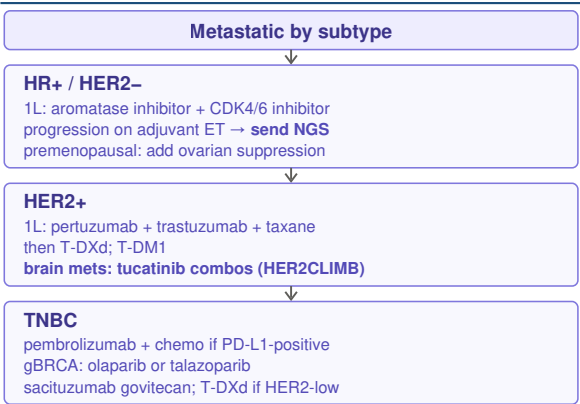
SURVEILLANCE

- H&P every 6–12 months × 5 years, then annually
- **Annual mammogram;** consider annual screening ultrasound if dense breast tissue, or CEM in lieu of mammogram + ultrasound
- **Annual gynecologic assessment** if on tamoxifen with a uterus
- **Baseline DEXA before starting an aromatase inhibitor** or after ovarian ablation, then every 1–2 years
- Assess and actively encourage **adherence to endocrine therapy**
- Lifestyle: alcohol <3 drinks/week, ≥150 min moderate exercise weekly, healthy BMI, Mediterranean diet
- DCIS: H&P + clinical breast exam q6 months × 5 years, mammogram q12 months

ADJUVANT ENDOCRINE THERAPY

- **Premenopausal:** tamoxifen 20 mg/d × **5–10 years** — 5 years cuts recurrence 40% and death 30%; extending to 10 years cuts recurrence a further 25% and mortality 29%
- Add **ovarian suppression** if high risk — goserelin 3.6 mg SQ q4wk, leuprolide 3.75 mg monthly, or oophorectomy (SOFT)
- **Postmenopausal:** aromatase inhibitor × 5–10 years — anastrozole 1 mg, letrozole 2.5 mg (non-steroidal), exemestane 25 mg (steroidal) daily. **Switch class if intolerant;** tamoxifen if AI contraindicated
- **Tamoxifen is a CYP2D6 prodrug** — avoid **paroxetine and fluoxetine** (also quinidine, cimetidine); citalopram and venlafaxine are acceptable. **CYP2D6 genotyping is not recommended**
- Menopause = prior bilateral oophorectomy, age ≥60, or <60 with amenorrhea ≥12 months off chemo/tamoxifen/ovarian suppression *and* postmenopausal FSH + estradiol

METASTATIC PATHWAY



BIOMARKER → THERAPY

Alteration	Therapy
Germline BRCA1/2	Olaparib or talazoparib
PD-L1-positive TNBC	Pembrolizumab + chemotherapy
PIK3CA	Inavolisib + palbociclib + fulvestrant · alpelisib · capivasertib
AKT1 / PTEN	Capivasertib + fulvestrant
ESR1	Elacestrant
HER2-low or ultralow	Trastuzumab deruxtecan (T-DXd)
MSI-H / dMMR	Pembrolizumab, dostarlimab
TMB ≥10 mut/Mb	Pembrolizumab
NTRK fusion	Larotrectinib, entrectinib, repotrectinib
RET	Selpercatinib

METASTATIC HR+/HER2– SEQUENCING

- **First line** — de novo metastatic, or progression >12 months after endocrine therapy: **aromatase inhibitor + CDK4/6 inhibitor**
- **Progression on adjuvant ET, or relapse within 12 months of it → check NGS.** PIK3CA-mutant: inavolisib + palbociclib + fulvestrant. Wild-type: fulvestrant 500 mg IM d0, 14, 28 then q28d + a CDK4/6 inhibitor
- **Progression on CDK4/6i + ET → switch CDK4/6 inhibitor** — ribociclib with alternative ET (MAINTAIN) or abemaciclib + fulvestrant (postMONARCH)
- Endocrine-resistant: **everolimus 10 mg + exemestane 25 mg daily**, then chemotherapy
- **ADCs:** T-DXd if HER2-low or ultralow · sacituzumab govitecan · datopotamab deruxtecan

SURGICAL & RADIATION PEARLS

- **Inflammatory breast cancer:** staging work-up → neoadjuvant chemotherapy → modified radical mastectomy → postoperative chest wall + regional nodal radiation — **even with a pathologic complete response.** Delayed reconstruction; BCS contraindicated; ALND indicated
- **Relative contraindications to BCS radiation:** Li-Fraumeni, active connective tissue disease involving skin (scleroderma, lupus), prior chest radiation, large tumor-to-breast-volume ratio
- **Reconstruction with planned radiation:** preferred is immediate tissue expander → expansion → radiation → exchange to implant, or delayed autologous
- Nipple-sparing mastectomy: ischemia and necrosis risk with high BMI, large or ptotic breasts, prior radiation or breast surgery
- **Lymphedema** ~5% after SLNB, highest after ALND + radiation — screen at visits; consider LYMPHA at nodal dissection; class I compression sleeve; PT referral
- **BIA-ALCL** — associated with textured implants; very rare; diagnosed on cytology; requires complete resection of capsule, implant and any mass